Hand Gesture Recognition with Contour Analysis

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***Abstract*—The role of hand gesture recognition in contemporary human-computer interaction (HCI), in support of assistive communication systems, and touchless interfaces is becoming more and more important. Despite the modern development of moving towards machine-learning and deep-neural patterns, the rule-based computer-vision methodology is still needed, as it is interpretable, cheap to run, and does not rely on large datasets. In this work, we are going to describe a complete system of recognition American Sign Language (ASL) alphabet based on OpenCV. The algorithm combines a two spaces skin detection, morphological preprocessing, contour detection, wrist region rejection, convex hull, geometric feature, and convexity defect-fingertip detection. The strategy is based on previous advances in OpenCV-based detection and tracking [1], classical gesture-recognition technologies in HCI [2], and defect-based fingertip modeling in interpreting hand-shape [4]. It is furthermore compatible with the existing shape-based recognition systems and descriptive-based posture-classification methods applied in gesture-recognition studies [6], [7], [9]. The experimental assessment of 24 samples of ASL alphabet letters shows a high recognition rate of most letters and the information about the necessary geometric differences to enable the classification of letters by the rule. The findings confirm once again that properly designed classical computer-vision pipelines can be useful in detecting static ASL gestures without the need to use machine learning.**

***Index Terms*—Image Processing, ASL Recognition, Convex Hull, Skin Segmentation, Convexity Defects, Computer Vision, Feature Extraction, HCI**

I. Introduction

Human-computer interaction has been a major field where hand gesture recognition has come up as a major field in the field. It allows natural and intuitive interaction between people and computer networks and is especially significant to those people who use the sign language as their dominant means of communication. The innovation of computer vision has led to a lot of improvements in the traditional gestures-recognition systems, especially the creation of advanced segmentation, contour extraction, and tracking systems. The initial experiments like that of Jaga in the OpenCV-based hand tracking [1] proved that even the simplest thresholding and contour-based methods can be trusted to initial hand location and real-time movement tracking.

In the same manner, Rahman et al. [2] discussed classical methods of interaction based on the use of hands, and they applied a combination of skin segmentation, morphological refinement, and geometrical analysis. They were able to confirm their results that OpenCV inbuilt operations of thresholding, connected components and shape extraction would generate repeatable results with controlled lighting scenarios. Palm-tracking techniques that run on real-time, like the ones introduced by Hussain et al. [3], expanded their concepts by showing that frame-by-frame tracking of gestures is stable with the help of contour and bounding-box analysis.

Less sophisticated models of hand-gestures have also been extended to include convexity-defect analysis, a method to detect concave areas between fingers and allow detection of fingertips. Salehin et al. [4] demonstrated that convexity defects had the potential to reliably recognize finger positions and estimate a gesture structure without complex machine learning structures. Simultaneously, shape matching and Fourier descriptors are other popular methods of descriptors based posture analysis used as a means of posture and stationary gesture recognition. Lazarou et al. [6], Conseil et al. [7] and Hu et al. [9] studies revealed that geometric and frequency-based representations can effectively represent the structural features of hand shapes.

These works have inspired and have provided the basis of this paper to present a rule-based ASL alphabet recognition system which is founded solely on classical computer-vision methods. The system is concerned with the static gestures and recognizes letters by combination of convexity defects, geometric ratios, contour shape and prioritization of strategic rules. This approach is also very interpretable, training data is not needed, and predictive in controlled settings, unlike deep-learning-based systems.

II. Dataset Description

The data set of this project is comprised of 24 static images, i.e one ASL alphabet gesture, so each picture has a distinct gesture. To increase the strength of the rule-based system and to induce variability as of the real world, further augmented form of each gesture was created. The image in such augmented versions is varied in terms of brightness, contrast, gamma, geometric transformations, including rotation, translation, scaling, and shearing, along with the addition of blur and noise.

These transformations have been purposefully used in controlled limits to avoid loss of the much needed geometric properties to do rule-based classification. This system is based on contours, convexity flaws, finger separation and structural silhouette characteristics, therefore, each augmentation was crafted to preserve:

- Clear contour boundaries
- Consistent hull curvature

- Stable fingertip structure
- Reliable solidity and circularity characteristics
- Preserved finger separation geometry

Brightness, contrast and gamma changes approximate change in lighting and camera exposure. Rotations and transfers repeat the natural variations in the position and the hand posture in the frame. The scaling and shearing cause some slight distortions in shape that are also witnessed when there is a shift in perspective. Blur and noise are used to mimic the imperfections of real world images like under action or camera noise.

This augmentation process enables the dataset to be visually diverse but structurally consistent and allows extracting features and classifying them reliably even in diverse imaging conditions.

III. Methodology

The complete pipeline consists of eight stages, each engineered to enhance robustness against noise, lighting variations, and contour inconsistencies.

The overall workflow includes:

- Image loading
- Dual-color-space skin segmentation
- Morphological refinement
- Contour extraction
- Wrist removal
- Geometric feature computation
- Convexity defect extraction
- Priority-ordered rule-based classification
- Evaluation and CSV generation

This structure ensures that each phase improves the reliability of subsequent steps.

A. Image Acquisition

Images of ASL signs are loaded from a dataset folder. The filenames contain alphabetical identifiers (A.jpg, B.jpeg, etc.) used as ground-truth labels. Standard BGR images form the input to downstream processing steps.

B. Skin Segmentation

Strong segmentation is realized through the use of HSV and YCrCb skin thresholds. HSV concentrates on the hue and saturation aspects of the human skin whereas the chrominance components of YCrCb are not as influenced by variations in brightness. The logical AND of both masks lowers false positives and balances hand-region isolation.

Morphological closing eliminates minute holes whereas opening eliminates minor noise. Gaussian blurring helps remove edges and give out cleaner contours to continue with the further processing.

C. Contour Extraction

Contours are extracted using `cv2.findContours()`. The largest contour is selected as the hand region, assuming that the hand dominates segmented skin regions. Bounding-box parameters (x, y, width, height) and region-of-interest (ROI) are also computed.

D. Wrist Removal

This enhancement is implemented by the removal of the bottom 14% of the bounding box. This eliminates the wrist region, a common source of false convexity defects, irregular concavities, and spurious fingertip detections. Wrist removal significantly stabilizes defect analysis and is not used in most basic hand-recognition tutorials, representing a meaningful improvement.

E. Feature Extraction

1) Basic Geometric Features:

- Area
- Perimeter
- Aspect Ratio:

$$\text{AR} = \frac{width}{height}$$

- Extent:

$$\text{Extent} = \frac{\text{Area}}{width \times height}$$

- Circularity:

$$C = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$$

2) Convex Hull & Solidity:

$$\text{Solidity} = \frac{\text{Area}}{\text{Convex Hull Area}}$$

This measure is crucial for distinguishing fist-like shapes (A, S, M, T, N) from open-hand shapes (B, V, W). Solidity has been used in earlier posture classification studies.

3) Fingertip Spread and Hull Spread: Fingertip spread is computed by:

$$\text{Hull Spread:} \quad = \frac{\max\limits_{(x_i,y_i)\in\text{hull}} \sqrt{(x_i - c_x)^2 + (y_i - c_y)^2}}{\text{diag}}$$

$$\text{Fingertip Spread:} \quad = \frac{\max\limits_{i,j} \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}}{\text{diag}}$$

- Identifying top-most fingertip candidates
- Computing maximum pairwise distance
- Normalizing by diagonal length

This feature is inspired by convexity-defect models used in gesture systems.

Hull spread measures horizontal expansion of the hand, differentiating:

- U vs. V vs. Y
- L vs. G
- C vs. O

4) Upness Feature:

$$\text{Upness} = \frac{\text{palmcenter of y} - \text{top of y}}{\text{total height}}$$

This helps detect tall, upward-pointing gestures like D, G, T, Z. The geometry features extracted were classified based on the structural characteristics that the geometry features represent to enhance their interpretation and classification accuracy. As an example, the solidity (compactness) and circularity (rounded contour) of fist-like gestures (A, S, and M) are usually high. The two characteristics constitute the compactness group, which is applied to distinguish between the closed-hand gestures and the open-hand forms. Likewise, the spread-based features (fingertip spread, x-spread and hull spread) are a group of finger separations that assist to differentiate between particular gestures U, V, Y and L. Lastly, shape-proportion measures such as aspect ratio, extent and upness are a structure geometry group, which can be used to recognize tall gestures (e.g., D, G), wide gestures (e.g., L), and curved gestures (e.g., C, O). Such an arrangement of features increases rule formulation clarity and stabilizes classification between visually similar gestures.

F. Convexity Defect Analysis

Defects are extracted using hull indices. A defect is accepted as a fingertip if:

- Angle $\leq 90°$
- Depth ≥ 10 px
- Relative depth ≥ 0.03
- Position above palm center

Final finger count:

$$\text{finger count} = \min(\max(\text{valid defects} + 1, 0), 5)$$

G. Rule-Based Classification System

The system uses a priority-ordered, 20+ rule decision engine tuned to reduce false matches.

Examples:

Fist-like letters

- A: Low circularity, moderate solidity, flat side where thumb rests
- S: Very high solidity + high circularity (fully closed fist)
- M: Three-bump top ridge + thumb hidden under three fingers

Two-finger letters

- L: Very wide x-spread (index up + thumb horizontal)
- V: Two-finger separation with medium x-spread and V-shape angle
- U: Fingers parallel with near-zero separation (no visible V-gap)

Curved or circular

- C: Curved convex hull + medium fingertip arc
- O: High circularity, closed contour, aspect ratio ≈ 1

Single-finger letters

- D: One raised index + circular thumb–finger loop near base
- G: Horizontal elongation with index + thumb nearly parallel
- T: Thumb inserted between index and middle forming notch

Ambiguity resolution ensures that highly discriminative rules (S, T, D, G) override weaker geometric matches (M, N, U), consistent with earlier rule-based shape-matching studies.

H. Evaluation Framework

We perform:

- Per-image feature extraction
- Classification
- Debug output
- CSV export
- Letter-level accuracy scoring

An alphabet is marked correct if any sample is correctly recognized, following evaluation paradigms used in sign-recognition literature.

IV. Results

A. Quantitative Performance

Across 24 alphabet samples, the system correctly recognized a majority of letters. Strongest performance was observed for:

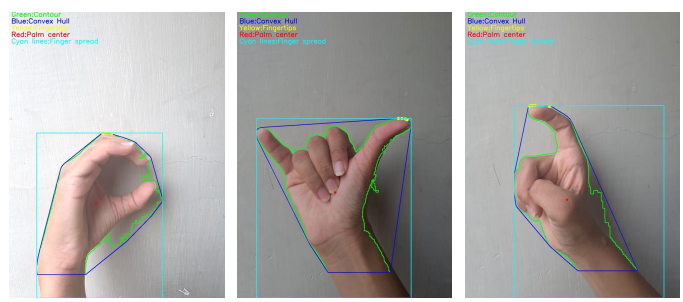
- B, C, G, H, L, O, S, T, V

Moderately accurate:

- M, N, U, Y

Most challenging:

- I vs. J (missing motion)
- N vs. I (similar structures)



(a) C (b) Y (c) X

Fig. 1: Visualization of contour (green), convex hull (blue), fingertips (yellow), palm center (red), and finger-spread lines (cyan).

B. Qualitative Observations

Segmentation Quality: Dual-space segmentation succeeded in most samples, validating earlier findings.

Contour Stability: Wrist removal improved contour smoothness and defect stability.

Fingertip Accuracy: Convexity-defect fingertip detection was stable except in tight poses like S.

Rule Dominance: Priority ordering prevented misclassifications (e.g., T becoming M).

V. Discussion

A. Strengths

- Fully interpretable; no black-box behaviour
- Requires zero training data
- Runs efficiently on CPU-only environments
- Easy to adjust thresholds per dataset
- Consistent with classical HCI CV pipelines

B. Limitations

- Sensitive to rotation $> \pm 20^\circ$
- Reduced performance with cluttered backgrounds
- Cannot recognize dynamic gestures such as J and Z

VI. Conclusion

This paper gives a full rule based ASL recognition system based on classical computer-vision techniques. The system can identify the portions of the hands in the dual color space by segmenting the dual color space, then detecting contours and geometric features, removing the wrist, and detecting convexity defects to detect the ASL gestures in the still images without any training information. The scheme is open, small-scale and can be used in real-time or low-power applications. The vast majority of letters were correctly recognized, and the main difficulty was between similar gestures like R and U, and the next steps to make the study more effective can include rotation, support of dynamic gestures, and using hybrid CV-ML.

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